

Oligool: Project Overview

Author: Mgr. Vojtěch Rejtar **Date:** February 24, 2026 **Project:** Oligool

Oligool is a native desktop application designed to streamline the design, alignment, and analysis of genetic sequences and custom oligos for molecular biologists.

Features and Capabilities

1. Dual-Source Search & Fetch

- Seamless toggling between NCBI and Ensembl data sources for finding genes and transcripts.

Smart filtering with persistent search parameters (E-value, Identity %, Organism) that survive application restarts.

Interactive MSA Viewer

- High-performance Multiple Sequence Alignment powered by MAFFT.

2D Navigation via an interactive minimap for scrubbing through massive alignments, highlighting conservation patterns, and identifying variations.

"Oligize!" Design

- Precision splitting of genomic regions into two contiguous oligos with exact control over shift and lengths.

Live integration with IDT OligoAnalyzer to evaluate hairpin formation and self-dimerization in real time (Delta G).

"Primerize!" Schematic

- High-fidelity SVG visual assembly of your molecular design, accurately representing Forward and Reverse Primer Binding Sites (PBS) and TAG sequences.
 - Seq Mode:** Toggle to a high-detail view displaying base-by-base lettering along the schematics architecture.
 - Persistence:** Local storage of user TAGs, PBS sequences, and design preferences.
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Typical Workflow

1. Launch the Application

- macOS:** Run the standalone Mac bundle `Oligool.app`.
- Windows:** Execute the single-file executable `Oligool.exe`.

All credentials (e.g., NCBI Key, IDT API authentication) and design configurations are securely stored locally via `localStorage`.

Search and Retrieve Sequences

- Toggle between NCBI and Ensembl.

Enter your target gene name or accession ID to retrieve annotated sequence data.

Sequence Alignment (Optional)

- Upload multiple sequences and utilize the MAFFT-powered MSA Viewer to align them.

Use the 2D minimap to identify conserved regions.

Oligo Design ("Oligize!")

12. Select a genomic region of interest.
13. Define your desired shift and oligo lengths to split the region into two contiguous oligos.

Review the live IDT OligoAnalyzer real-time Delta G values to avoid hairpins and self-dimerization.

Visual Assembly ("Primerize!")

16. Navigate to the schematic view to visually assemble your molecular design.
17. Input your custom TAGs, Forward PBS, and Reverse PBS.
18. Enable **Seq Mode** to verify the base-by-base construct architecture.
19. Use the generated sequence constructs for your downstream experimental workflows.